

Unit 2, Lesson 4: Converting Between Repeating Decimals and Fractions

Benchmark

MA.7.NSO.1.2 Rewrite rational numbers in different but equivalent forms including fractions, mixed numbers, repeating decimals, and percentages to solve mathematical and real-world problems.

Learning Target

- ☐ I can rewrite rational numbers in different but equivalent forms, including repeating decimals.

2.4.1 Warm-Up: Error Analysis

1. Isaiah used a scientific calculator to find the decimal equivalent of $\frac{16}{6}$. The calculator screen displayed "2.666666667". Isaiah wrote the following equation on his paper:

$$\frac{16}{6} = 2.666666667$$

- a. Explain why Isaiah's equation is incorrect.
- b. What advice would you give Isaiah when using a calculator to find decimal equivalents in the future?

2.4.2 Guided Instruction: Rewriting Repeating Decimals as Fractions

When given a repeating decimal, equations can be used to find its equivalent value represented as a fraction.

1. Consider the decimal $0.0\overline{6}$. Some fraction is equivalent to $0.0\overline{6}$, which can be represented by the equation, $x = 0.0\overline{6}$.
 - a. If $x = 0.0\overline{6}$, complete the step shown to determine what value is equivalent to $10x$.

$$10 \cdot x = 0.0\overline{6} \cdot 10$$

$$10x = \boxed{}$$

- b. Determine what value is equivalent to $100x$.

$$\underline{\hspace{1cm}} \cdot x = 0.0\overline{6} \cdot \underline{\hspace{1cm}}$$

$$100x = \boxed{}$$

- c. These relationships can be used to find the value of x represented as a fraction. Complete the steps shown.

$$100x - 10x = \boxed{} - \boxed{}$$

$$\underline{\hspace{1cm}}x = \underline{\hspace{1cm}}$$

$$x = \frac{\boxed{}}{\boxed{}}$$

- d. Rewrite the solution as a fraction in simplest form.

2. Consider the repeating decimal $0.\overline{36}$.

- a. If $x = 0.\overline{36}$, complete each equation.

$$x = 0.\overline{36}$$

$$10x = \underline{\hspace{2cm}}$$

$$100x = \underline{\hspace{2cm}}$$

- b. Which two values, when subtracted, would result in the repeating portion of the decimal being eliminated?
- c. Use two of the relationships represented in the equations in Part A to complete the steps to find the fraction equivalent of $0.\overline{36}$.

$$\underline{\hspace{2cm}} - \underline{\hspace{2cm}} = \underline{\hspace{2cm}} - \underline{\hspace{2cm}}$$

$$\underline{\hspace{2cm}}x = \underline{\hspace{2cm}}$$

$$x = \frac{\boxed{\hspace{1cm}}}{\boxed{\hspace{1cm}}}$$

- d. Rewrite the solution as a fraction in simplest form, if possible.

2.4.3 Guided Practice: Rewriting Repeating Decimals as Fractions

1. Consider the repeating decimal $0.3\overline{8}$.

a. If $x = 0.3\overline{8}$, complete each equation.

$$x = 0.3\overline{8}$$

$$10x = \underline{\hspace{2cm}}$$

$$100x = \underline{\hspace{2cm}}$$

b. Which two values, when subtracted, would result in the repeating portion of the decimal being eliminated?

c. Use two of the relationships represented in the equations in Part A to complete the steps to find the fraction equivalent of $0.3\overline{8}$.

$$\underline{\hspace{2cm}} - \underline{\hspace{2cm}} = \underline{\hspace{2cm}} - \underline{\hspace{2cm}}$$

$$\underline{\hspace{2cm}}x = \underline{\hspace{2cm}}$$

$$x = \frac{\boxed{\hspace{1cm}}}{\boxed{\hspace{1cm}}}$$

d. Rewrite the solution as a fraction in simplest form, if possible.

e. Write the fraction or mixed number that is equivalent to $14.3\overline{8}$.

2.4.4 Collaboration: Fractions, Decimals, and Percentages

Work with your partner to complete the following.

1. In a standard deck of Uno cards, there are 24 action cards that include “Skip,” “Draw Two,” and “Reverse.” There are 108 cards in an Uno deck.

a. What fraction of the Uno cards are action cards?

b. Express your answer to part a as a decimal and a percentage.

Decimal	Percent

- c. Discuss with your partner why 22% would be an incorrect equivalent value and summarize your discussion in the space below.

2. Aliyah played a game using one standard die. There is a $33.\bar{3}\%$ chance of winning the game on any roll.

a. Express the chances of winning the game as a fraction and a decimal.

Decimal	Fraction

- b. Which representation – decimal, fraction, or percentage – is the easiest way to describe the chance of a winning roll?

- c. What are the possible results in the sample space when rolling a standard six-sided die?
- d. Discuss with your partner what you think the rules of Aliyah's game might be. Then, complete the statement with one possibility.

To win the game, Aliyah must roll _____

2.4.5 Your Turn!

1. In a package of Skittles, Carmela found that 21% of them were red and $0.1\bar{6}$ were green.
 - a. What fraction of the Skittles was red?
 - b. What fraction of the Skittles was green?
 - c. She found 17 of the 60 skittles are purple. What percent of the Skittles are purple?

2.4.6 Wrap-Up: Ticket Out the Door

1. Complete the table.

Decimal	Percent	Fraction	Mixed Number, if Possible
$0.\overline{63}$			
	$12.\overline{12}\%$		
		$\frac{16}{3}$	
			$9\frac{5}{22}$

